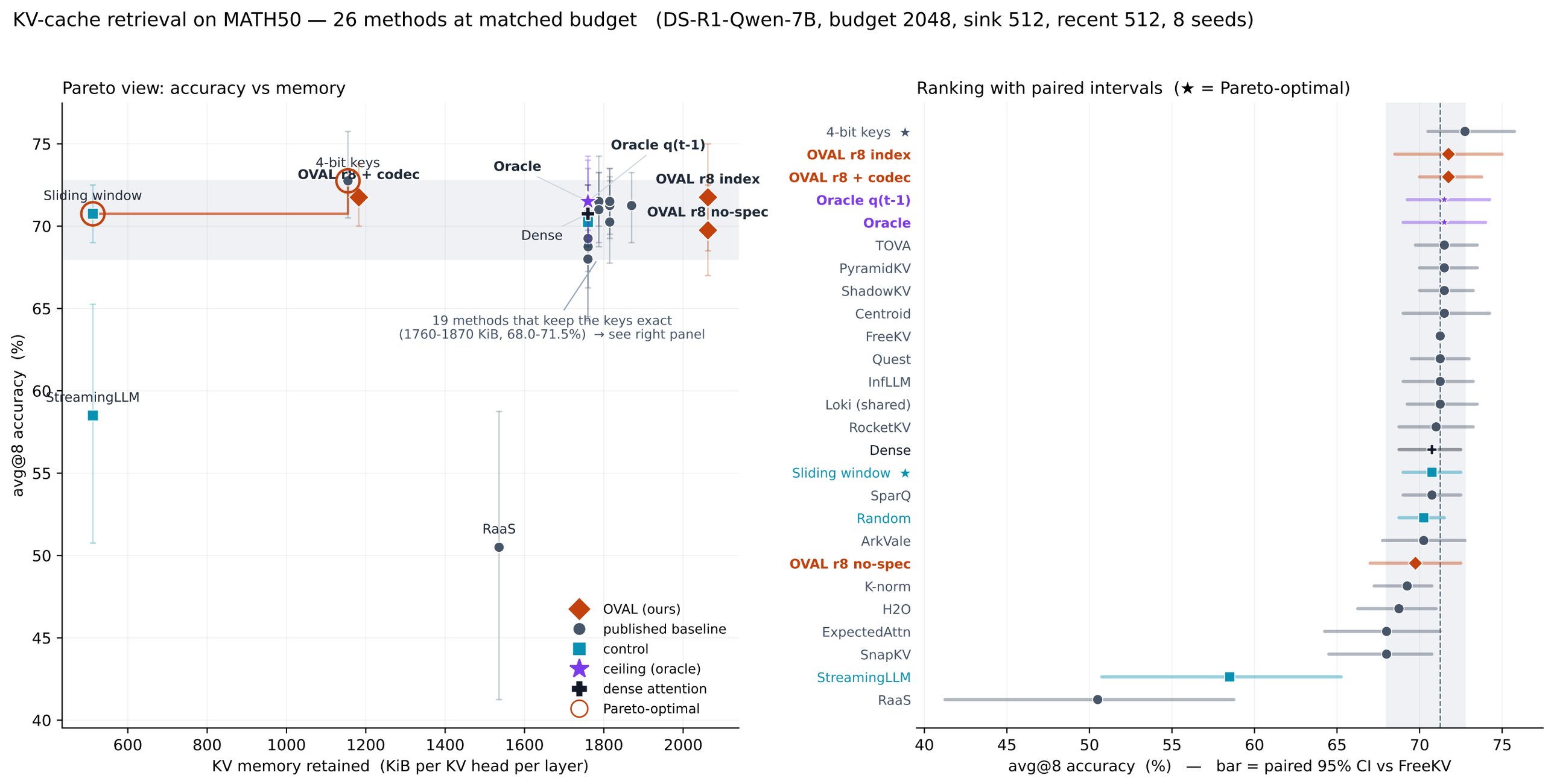
KV-cache retrieval on MATH50 — 26 methods at matched budget (DS-R1-Qwen-7B, budget 2048, sink 512, recent 512, 8 seeds)
Pareto view: accuracy vs memory
avg@8 accuracy (%)
KV memory retained (KiB per KV head per layer)
4-bit keys
OVAL r8 + codec
Oracle q(t-1)
Oracle
OVAL r8 index
OVAL r8 no-spec
Sliding window
Dense
StreamingLLM
RaaS
19 methods that keep the keys exact
(1760-1870 KiB, 68.0-71.5%) → see right panel
OVAL (ours)
published baseline
control
ceiling (oracle)
dense attention
Pareto-optimal
Ranking with paired intervals (★ = Pareto-optimal)
4-bit keys ★
OVAL r8 index
OVAL r8 + codec
Oracle q(t-1)
Oracle
TOVA
PyramidKV
ShadowKV
Centroid
FreeKV
Quest
InfLLM
Loki (shared)
RocketKV
Dense
Sliding window ★
SparQ
Random
ArkVale
OVAL r8 no-spec
K-norm
H2O
ExpectedAttn
SnapKV
StreamingLLM
RaaS
avg@8 accuracy (%) — bar = paired 95% CI vs FreeKV
Both oracles sit inside the tie band: even with perfect knowledge of the true attention distribution there is nothing to win at this budget, because only ~20% of a
reasoning trace is ever compressed (onset at token 3072, mean sequence 3869). No ordering within the band is a result — the separable findings are the five arms below it.